IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science.

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Methodology Summary

- Data collection via the SpaceX REST API
- Web scraping
- Data wrangling
- Exploratory analysis with SQL
- Exploratory data analysis with visualization
- Interactive visual analysis with Folium
- Dashboards with Plotly Dash
- Predictive modeling (machine learning)

Results Summary

- Success rate trend over time
- Best-performing launch site
- Orbits with a 100% success rate
- Geographic advantage of the launch bases
- Accuracy of the machine learning models
- Best overall performing model
- Most influential factors for success

Introduction

SpaceX

- Space Exploration Technologies Corp. (SpaceX) is a private aerospace company founded in 2002 in the United States by Elon Musk, built around reusable rocket technology.
- Under the traditional approach the rocket was discarded after a single use, pushing costs to roughly US$165 million per launch.
- SpaceX learned to land the Falcon 9 first stage vertically on land or on drone ships, cutting the cost to roughly US$62 million.

Project Inputs

- Public launch data gathered through web scraping and the SpaceX REST API, explored and visualized with SQL and Python, then used to build machine learning prediction models.

Project Objectives

- Predict whether the first stage will land successfully
- Estimate launch costs
- Gain competitive advantage
- Identify patterns behind successful landings
- Build a reliable predictive model

Key Questions

- Does payload mass affect the landing?
- Does the target orbit change the outcome?
- Does the launch site make a difference?
- Is landing on ground safer than at sea?
- Does accumulated experience improve results?
- Which algorithm is the most accurate?

Methodology

Methodology Overview

- **Data Collection**
  - Historical launch records pulled from the SpaceX REST API and complemented with web scraping from Wikipedia.
- **Data Wrangling**
  - Falcon 9 launches filtered, missing payload values treated, and a binary landing outcome variable created.
- **Exploratory Analysis with SQL**
  - Queries run in Jupyter notebooks to surface success rates by site, orbit and year.
- **Visual Exploratory Analysis**
  - Bar, line and scatter charts built with matplotlib, seaborn and plotly to expose trends and correlations.
- **Geospatial Analysis**
  - Launch bases mapped with Folium to assess the influence of location on landing outcomes.
- **Interactive Dashboard**
  - A Plotly Dash application built to filter results by site, orbit and payload range.
- **Predictive Modeling**
  - Four classification algorithms trained, tuned and compared with scikit-learn.

Data Collection – SpaceX API

A four-step pipeline from API request to a parsed dataset

1

API Query

Request launch data from the SpaceX REST API endpoints.

2

Response Data

Receive the raw JSON payload with launch records.

3

Data Storage

Persist the raw response for reproducible access.

4

Data Parsing

Extract and normalize fields into a clean table.

Source: SpaceX REST API

Data Wrangling

Cleaning and shaping the launch records into a model-ready dataset

1

Handling Missing Values

Identify and treat gaps in payload mass and related fields.

2

Launch Count by Site

Count how many launches occurred at each launch site.

3

Orbit Analysis

Count the occurrences of each target orbit type.

4

Landing Success Indicator

Derive a binary class label from the mission outcome.

5

Missing Outcome Evaluation

Review outcomes that remain unknown after cleaning.

Source: SpaceX REST API

Exploratory Data Analysis with SQL

Querying the launch dataset directly to surface early patterns

Tool: SQL queries run in Jupyter notebooks

ACTIVITIES

Success Rate by Site

Compare landing success across each launch site.

Success Rate by Orbit

Rank orbit types by mission outcome.

Payload Mass by Outcome

Average payload mass for every result class.

Landings over Time

Track how success evolved year by year.

Source: SpaceX REST API

Tools and Technology Stack

- **Data Acquisition**
  - Python with requests and BeautifulSoup for the SpaceX REST API and Wikipedia web scraping.
- **Data Processing**
  - pandas and numpy for filtering, cleaning, feature creation and one-hot encoding.
- **Querying**
  - SQL executed inside Jupyter notebooks for aggregate analysis of the launch dataset.
- **Static Visualization**
  - matplotlib and seaborn for bar, line and scatter charts.
- **Geospatial Mapping**
  - Folium for interactive maps of the launch bases and landing zones.
- **Interactive Dashboard**
  - Plotly and Plotly Dash for a filterable dashboard of launch outcomes.
- **Machine Learning**
  - scikit-learn for model training, cross-validation, hyperparameter tuning and evaluation.

Exploratory Data Analysis with Visualization

Charting the cleaned dataset to expose trends, correlations and spread

Tools: Python — matplotlib, seaborn, plotly

ACTIVITIES

Bar Charts

Compare outcomes across sites, orbits and boosters.

Line Charts

Show how success improved over the years.

Scatter Plots

Reveal correlations between payload, orbit and result.

Distribution Views

Inspect how each variable spreads across the data.

Source: SpaceX REST API

Geospatial Analysis and Interactive Dashboard

Mapping the launch bases and making the results explorable

Tools: Folium for maps, Plotly Dash for the dashboard

ACTIVITIES

Site Mapping

Plot every launch base on an interactive map.

Spatial Context

Show the surroundings of each launch location.

Filter by Site

Drill into outcomes for a single launch base.

Filter by Orbit

Compare performance across target orbit types.

Source: SpaceX REST API

Model Setup and Feature Engineering

- **Target Variable**
  - Binary classification target: 1 = successful landing, 0 = failure.
- **Selected Features**
  - Payload mass, orbit type, launch site and flight number.
- **Preprocessing**
  - One-hot encoding for categorical variables and StandardScaler for numeric variables.
  - Dataset split into 80% training and 20% test.
- **Training and Tuning**
  - Four classifiers trained: Logistic Regression, SVM, Decision Tree and K-Nearest Neighbors.
  - K-fold cross-validation for robustness and GridSearchCV for hyperparameter optimization.
- **Evaluation Metrics**
  - Accuracy, precision, recall, F1-score and confusion matrix computed for every model.

Predictive Modeling with Machine Learning

From prepared features to the best-performing landing classifier

Tool: Python — scikit-learn

1 Prepare the Data

Binary target (1 = successful landing), key features, one-hot encoding, scaling and an 80/20 split.

2 Train the Models

Fit four classification algorithms on the training set with scikit-learn.

3 Tune and Validate

K-fold cross-validation plus GridSearchCV to find the best settings.

4 Evaluate and Choose

Accuracy, precision, recall, F1-score and confusion matrix pick the winner.

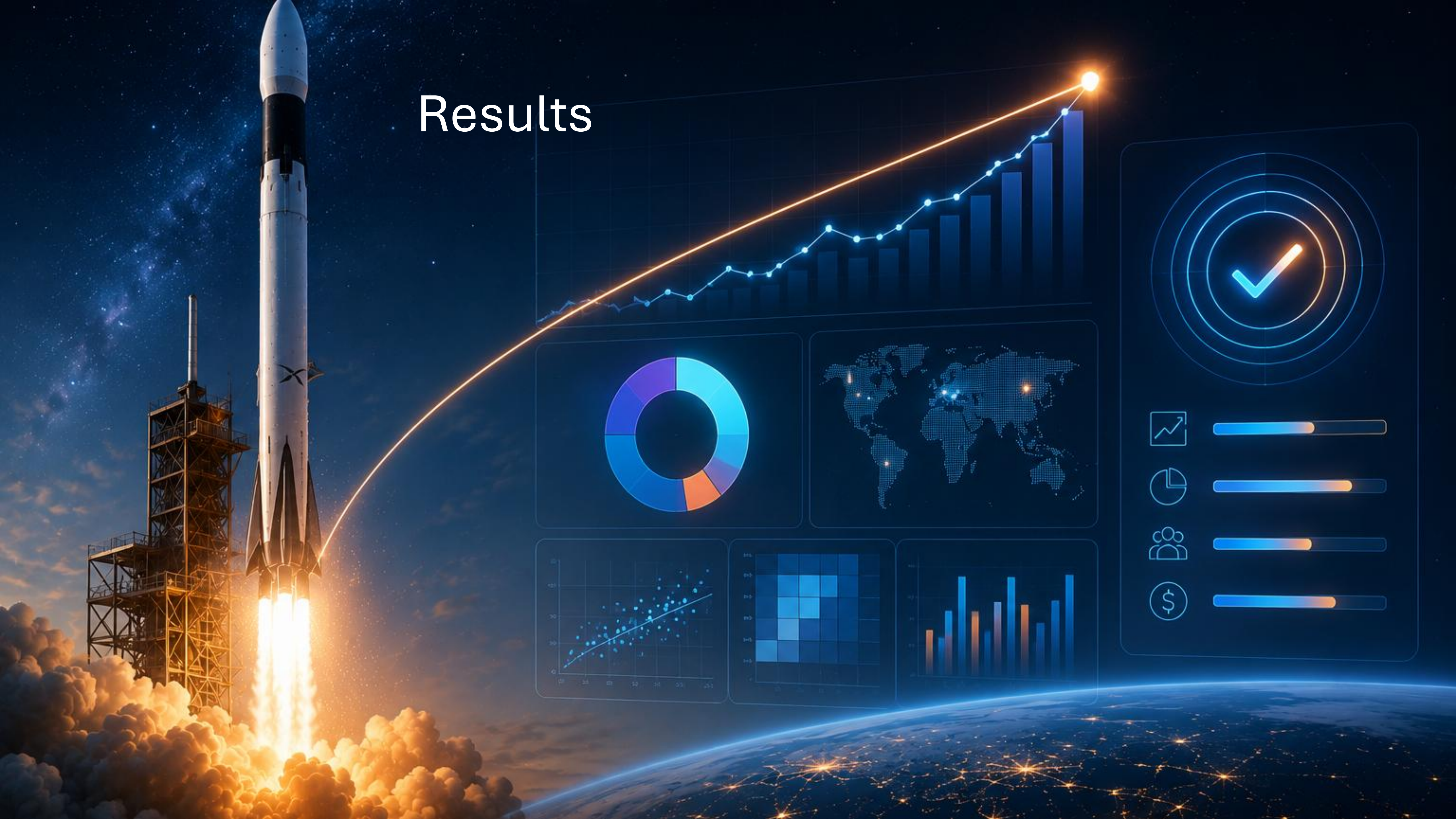
ALGORITHMS COMPARED

Logistic Regression · SVM · Decision Tree · KNN

All models above 90% accuracy

Source: SpaceX REST API

Results

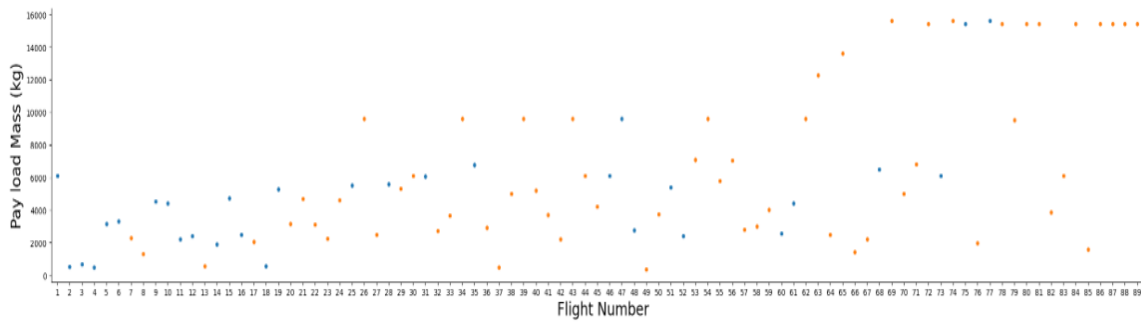


EDA with Visualization - Results

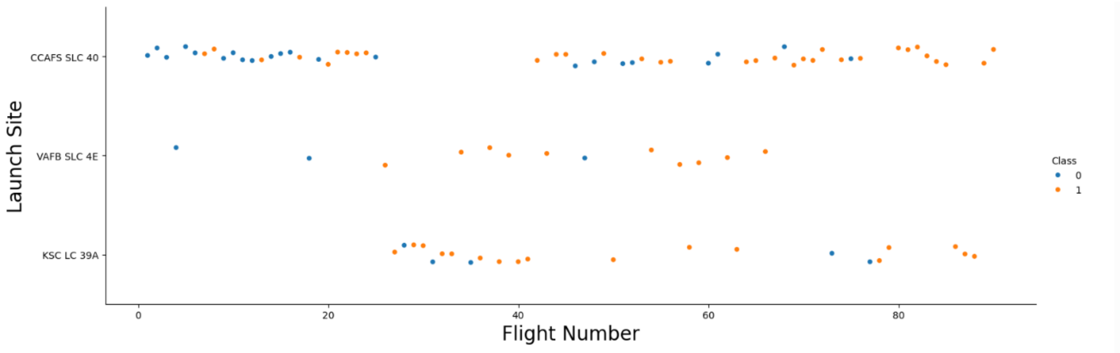
- **Trends over Time**
  - Line charts confirm a continuous improvement in landing success rate across the years.
- **Site and Orbit Performance**
  - Bar charts identify KSC LC-39A as the strongest launch site, with GEO, HEO and SSO orbits reaching a 100% success rate.
- **Payload Mass**
  - Scatter plots show that heavier payloads are associated with a clearly higher failure rate.
- **Geospatial View**
  - Folium maps colored by outcome confirm the bases are positioned near the coast and close to the Equator, which favors landing recovery.
- **Interactive Exploration**
  - The Plotly Dash dashboard allowed filtering by site, orbit and payload range, revealing non-linear patterns that static charts alone did not expose.

EDA with Visualization - Charts

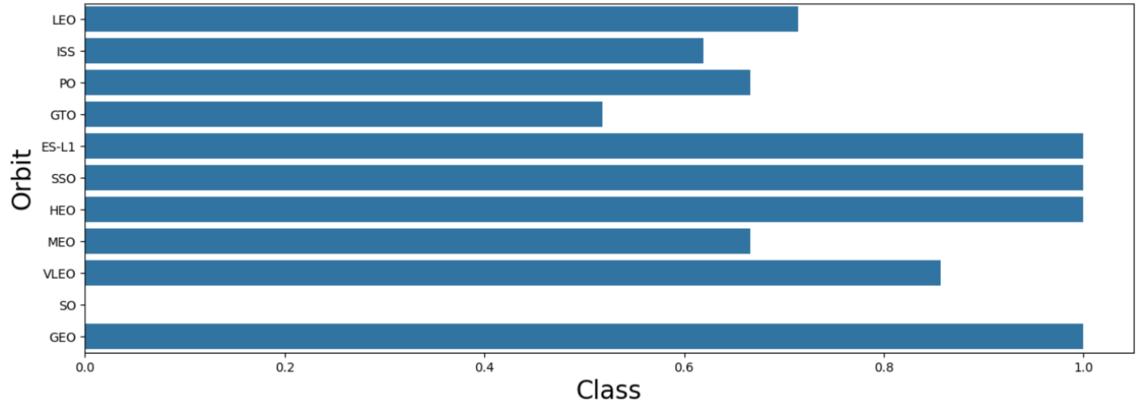
Payload mass vs flight number



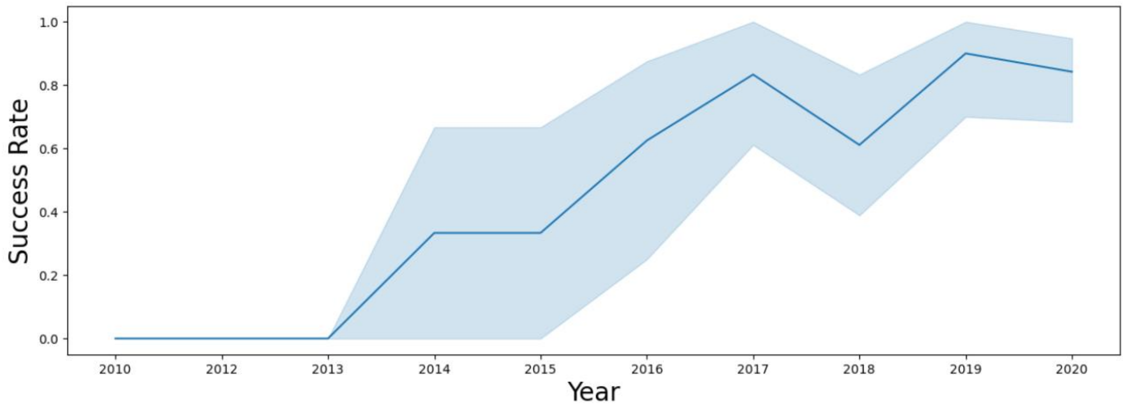
FlightNumber vs LaunchSite



Success rate by orbit type



Launch success yearly trend



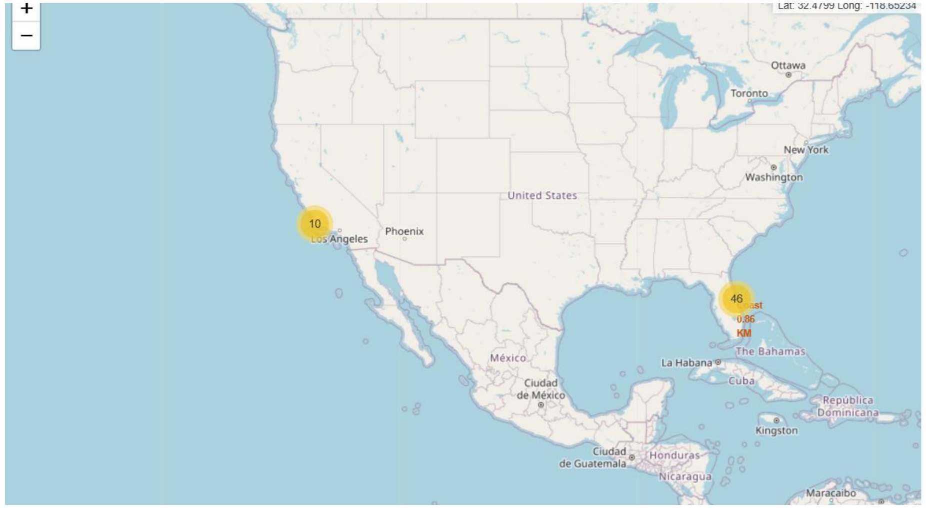
EDA with SQL - Results

- **Overall Success Rate**
  - Across the full Falcon 9 history the overall landing success rate exceeds 60%, providing the baseline for every later analysis.
- **By Launch Site**
  - GROUP BY queries show KSC LC-39A with the highest success rate, while CCAFS SLC-40 trails behind.
- **By Orbit Type**
  - GEO, HEO and SSO orbits reach 100% success, while GTO is markedly lower, indicating a harder landing profile.
- **By Year**
  - Success was low in 2015-2016 and rose sharply from 2018 onward.
- **Payload Mass Statistics**
  - Aggregate functions show the average payload of successful missions is lower than that of failed ones, confirming that heavier payloads reduce landing odds.

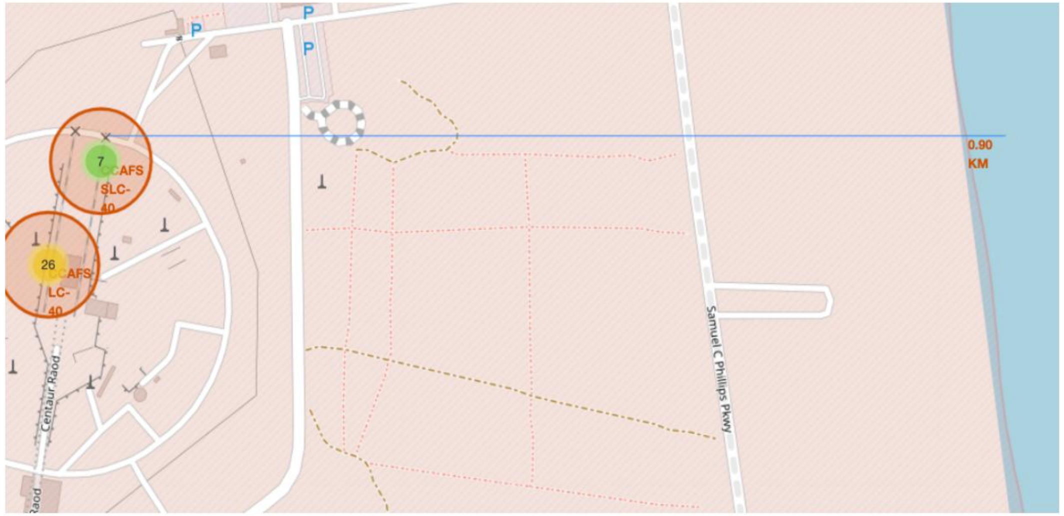
Interactive Map Results with Folium

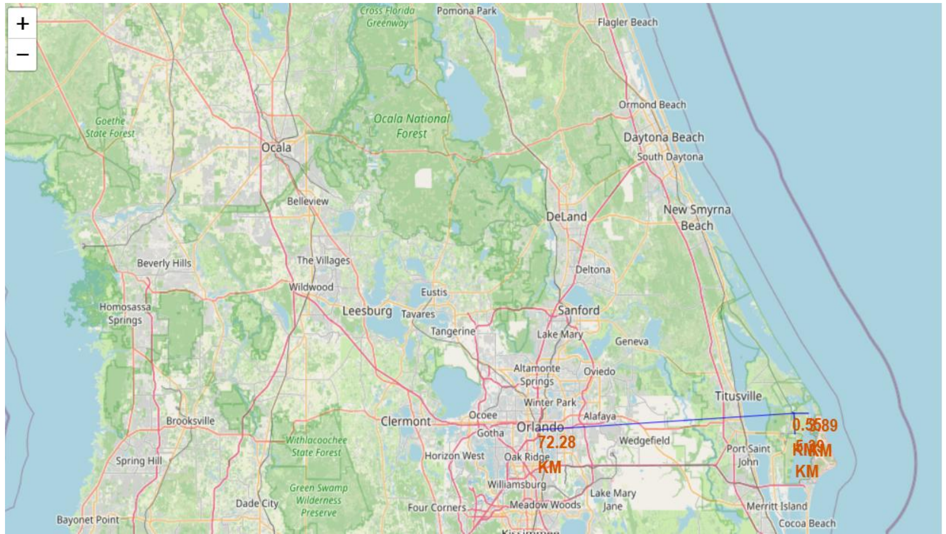
- **Geographic Distribution**
  - All Falcon 9 pads sit in Florida (CCAFS SLC-40 and KSC LC-39A) and California (VAFB SLC-4E), every one of them on the coast so trajectories run over the ocean.
- **Proximity to the Equator**
  - The Florida bases are significantly closer to the Equator than Vandenberg, which explains their preference for geostationary missions.
- **Performance by Site**
  - Markers colored green for success and red for failure show a far higher concentration of successes at KSC LC-39A, matching the SQL findings.
- **Landing Zones**
  - Radius circles drawn around each base confirm that most landing zones, including the drone ships, fall within recovery range of the coast.
- **Regional Pattern**
  - East coast launches show a higher overall success rate than west coast launches, reflecting mission profile and more mature infrastructure.

Interactive Map Results with Folium



Your updated map with distance line should look like the following screenshot:

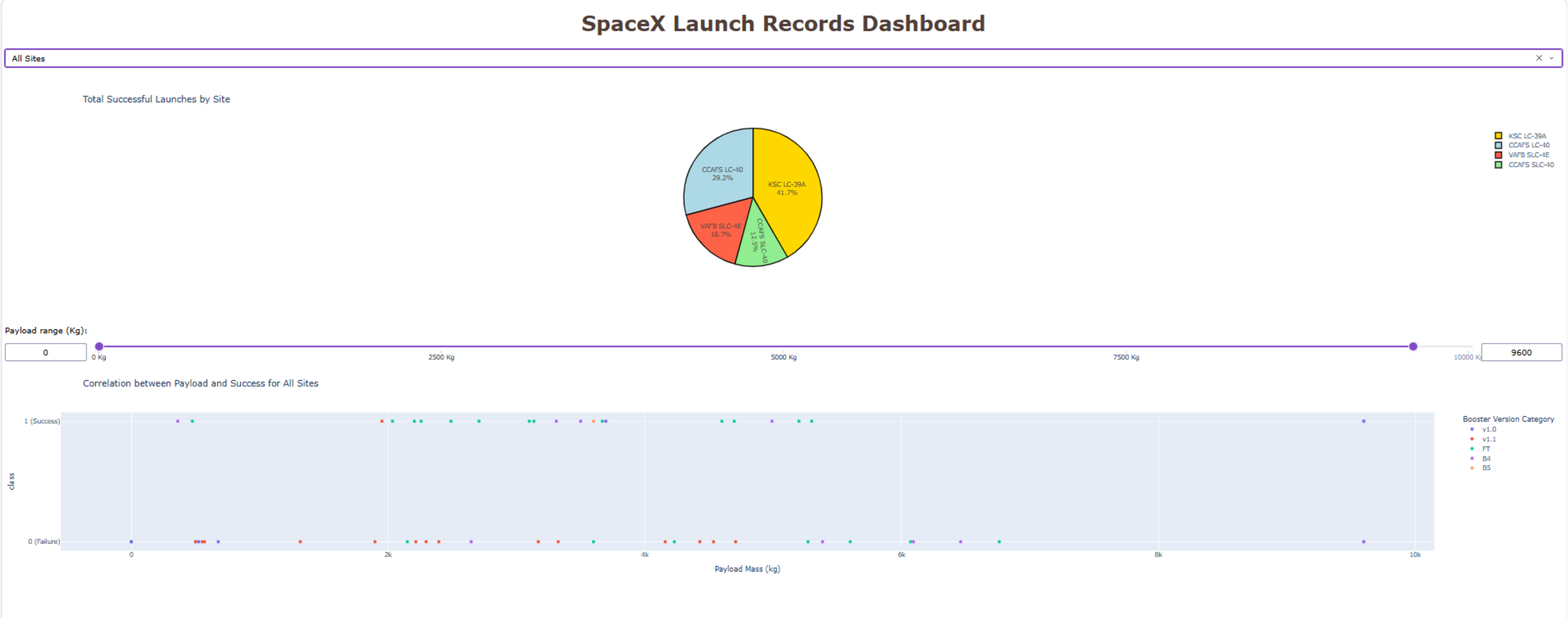




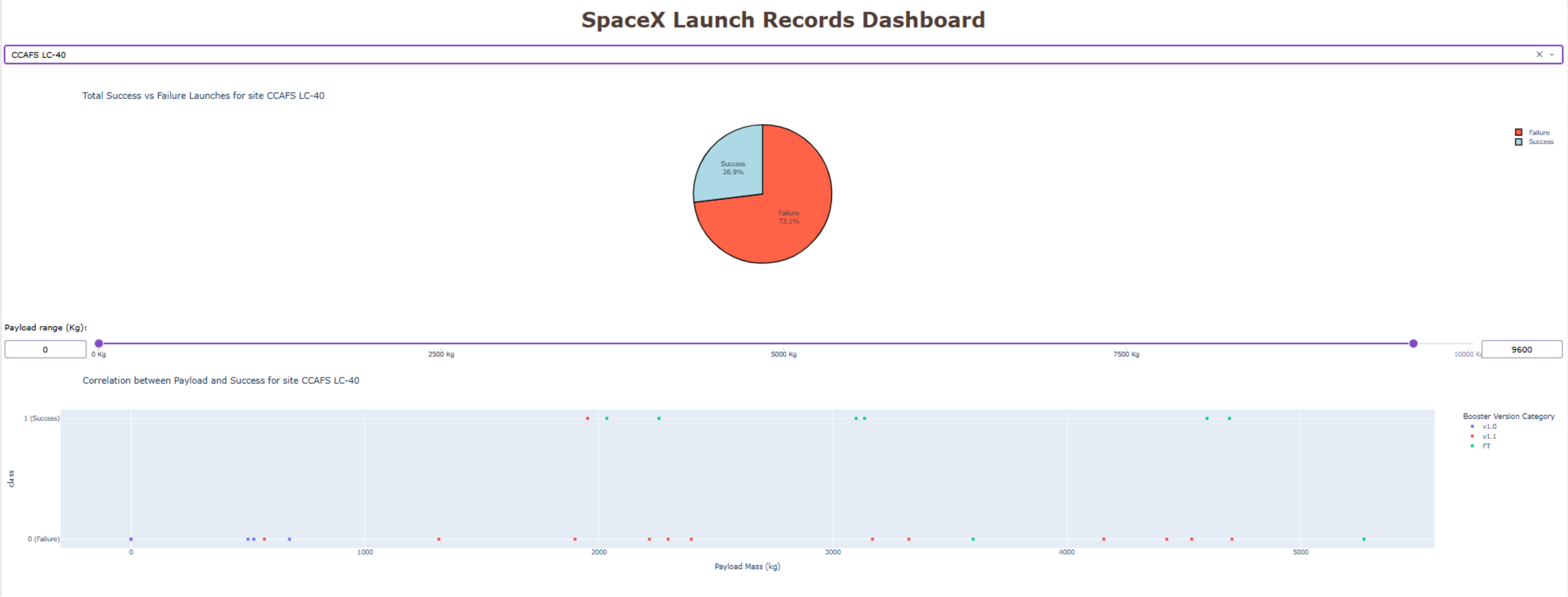
Plotly Dash Dashboard - Results

- **Success Rate by Site**
  - Switching between sites confirmed in real time that KSC LC-39A holds the highest success rate, above 80%, ahead of CCAFS SLC-40 and VAFB SLC-4E.
- **Payload Mass Correlation**
  - The payload slider showed success collapsing above 5,500 kg and exceeding 90% below 4,000 kg, confirming a strong negative relationship between weight and landing success.
- **Temporal Inflection Point**
  - Regardless of site or payload, success rates climb sharply from 2017 onward, indicating that technical improvements benefited every mission category.
- **Orbit Cross-Filtering**
  - GTO missions consistently underperform LEO, GEO, HEO and SSO. Crossing orbit with site shows KSC handling GTO with reasonable success while Vandenberg specializes in polar orbits.
- **Non-Linear Patterns**
  - Interactive scatter plots revealed a sweet spot between 1,500 kg and 4,000 kg where success is near-certain, with very low payload test missions also showing failures.
- **Integrated Decision View**
  - Combining every filter on one screen let an analyst instantly estimate the success odds of a specific mission profile, which static charts could not support.

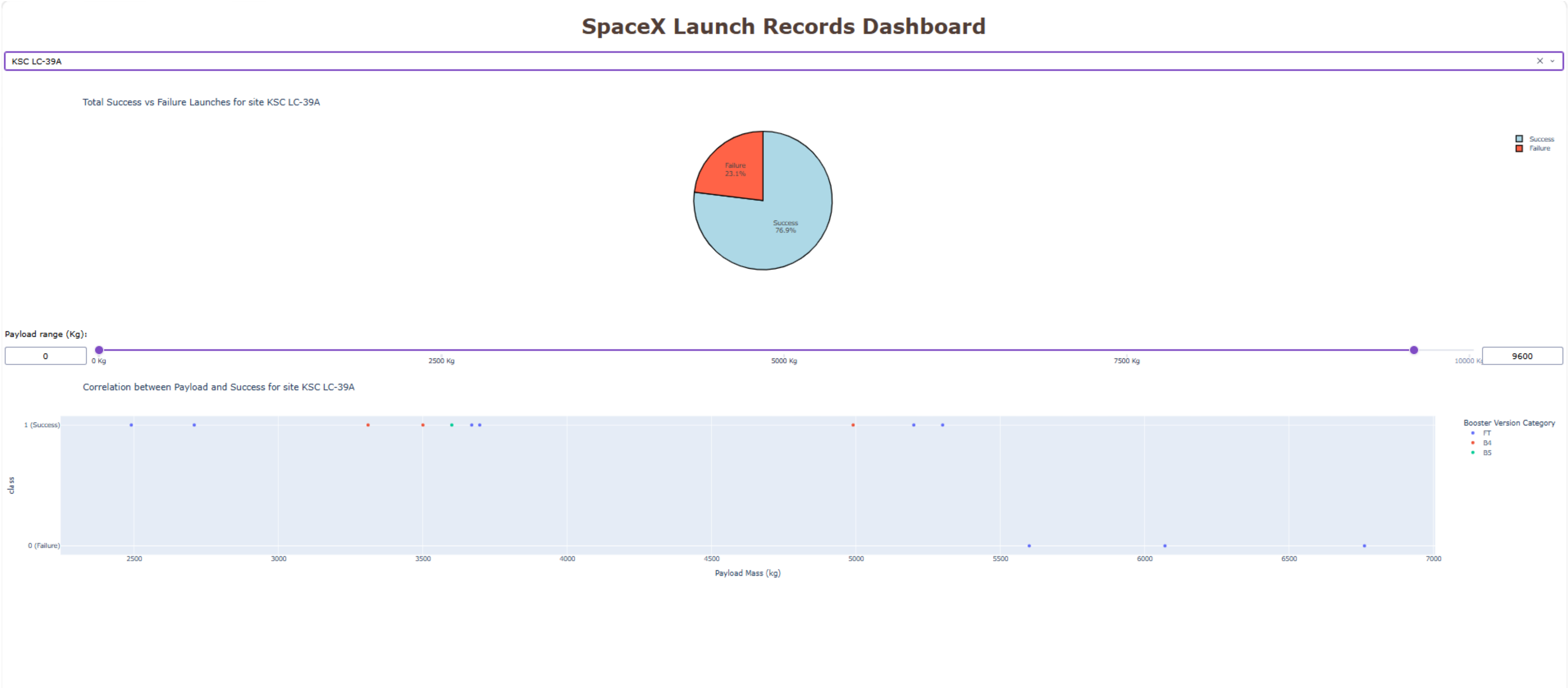
Plotly Dash Dashboard - Results



Plotly Dash Dashboard - Results



Plotly Dash Dashboard - Results



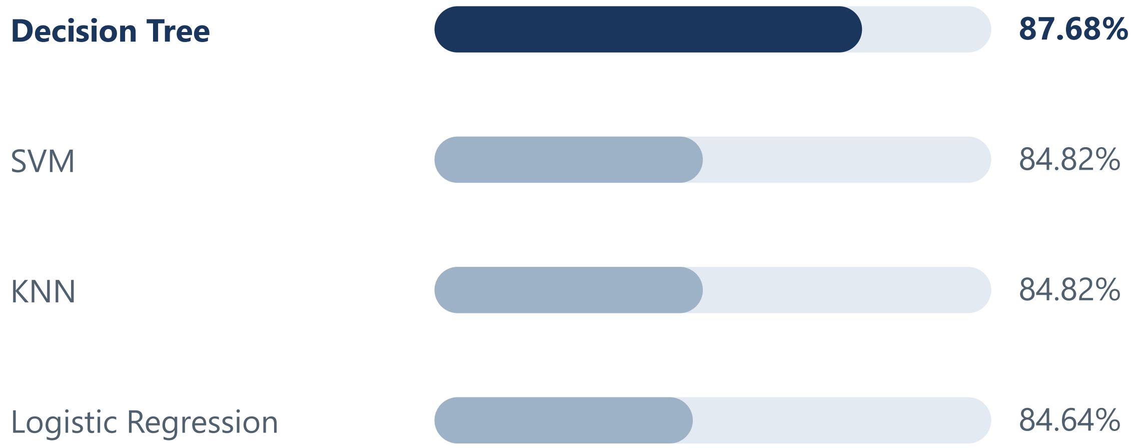
Predictive Analysis - Classification Results

- **Overall Accuracy**
  - All four models - Logistic Regression, SVM, Decision Tree and KNN - scored between 84.6% and 87.7%.
- **Selected Model**
  - Decision Tree, chosen for the best balance of high accuracy, F1-score around 0.86 and strong interpretability.
- **Most Important Factors**
  - Payload mass, over 30% importance - heavier payloads hurt landing success.
  - Orbit type, around 25% - GTO shows the most failures.
  - Flight number, around 20% - recent missions succeed more often.
- **Predicted Success Profile**
  - Payload under 4,000 kg, LEO or GEO orbit, launched from KSC in Florida after 2018 gives a landing probability above 95%.
- **Conclusion**
  - Landing outcomes can be predicted with high confidence, above 87.5% accuracy, using historical flight data alone.

Model Results

Accuracy comparison and confusion matrix of the final classifier

ACCURACY BY MODEL



Bar length shown on an 80–90% scale to make the gap visible

CONFUSION MATRIX

11 **True positives** Landings correctly predicted as successful	**1** **False positives** Predicted success, actually a failure
1 **False negatives** Predicted failure, actually a success	**5** **True negatives** Failures correctly predicted

16 of 18 test launches classified correctly

The Decision Tree leads by roughly 3 percentage points and balances both classes with few errors.

Source: SpaceX REST API

Key Findings

- **The Pipeline Worked End to End**
  - From API collection and web scraping through cleaning, transformation and machine learning, every stage ran as an integrated workflow that turned unstructured real-world data into decision-ready insight.
- **Critical Success Factors Identified**
  - Payload mass is the primary obstacle - heavier payloads sharply reduce landing odds.
  - Orbit type sets the difficulty - GTO is the most challenging, while LEO and GEO perform best.
  - KSC LC-39A outperforms the other sites thanks to its position closer to the Equator.
  - Success climbed from low levels in 2015 to above 80% in recent years as SpaceX iterated.
- **Predictive Modeling Delivered Robust Results**
  - The four classifiers all exceeded 84% accuracy, with the Decision Tree selected as the final model at roughly 87.7%. Historical data carries enough signal to estimate mission risk before launch.
- **Practical Value**
  - The results are directly applicable in the aerospace industry, letting engineers and managers price and de-risk a mission before committing to a launch.

Conclusion

What the SpaceX capstone project delivered

Falcon 9 first-stage landings can be predicted before launch with about 87.7% accuracy.

FINAL MODEL

Decision Tree

Best accuracy and the strongest balance between precision and generalization.

KEY DRIVERS

Payload, orbit, site

Payload mass, orbit type and launch location shape the odds of a successful landing.

VALUE DELIVERED

Risk known upfront

Mission risk can be estimated before launch, saving cost and improving safety.

Appendix

- SpaceX REST API - https://api.spacexdata.com/v4/launches/past
- Wikipedia web scraping - Falcon 9 and Falcon Heavy launch tables
- CSV datasets - spacex_launch_geo.csv, dataset_part_2.csv, dataset_part_3.csv